

M^5 Day 2: More numbers

Name: _____

1. State whether these statements are True or False. If False, give one example showing why.

- a. Every integer is also a real number.

- b. The product of two irrational numbers is irrational.

- c. If a is an integer, then $a^2 > a$.

- d. If $ab = ac$ for real numbers a, b, c , then it must follow that $b = c$.

- e. If a not rational, then a^2 is also not rational.

- f. The distance between any two real numbers must be positive.

2. Plot the following sets of real numbers on the number line:

a. The real numbers that are larger than 0 and less than or equal to 1.

b. The real numbers x such that $x < 3$.

c. The real numbers x that are at most distance 2 away from 7.

d. The real numbers x that are more than distance 1 away from 0.

e. The real numbers x that satisfy both $x < 4$ and $|x - 3| < 2$.

f. The real numbers x that satisfy either $x \geq -1$ or $|x + 7| < 1$.